

Scenario 1: Pretrained Model Predicts On

Context: You fine-tuned a pretrained YOLOv5 model on a custom dataset (detecting helmets). After training, it always predicts only one class.

Problem: Despite having multiple classes in the dataset, the model outputs only “helmet” for every object

Answer:

1. first check the annotation files and `data.yaml`.
2. then verify that the class IDs are correctly assigned, that `nc` matches the number of classes, and that the dataset is not severely imbalanced.
3. Should also visualize the training labels to ensure that the different classes are actually present. If the labels and configuration are correct, then check the training loss, class distribution, and inference confidence thresholds.

Scenario 2: Transfer Learning Model Trains Too Slowly on Small Dataset

Context: You are using a pretrained ResNet50 model for face mask classification (2 classes). You fine-tune the entire model on a small dataset (500 images).

Problem: Training takes too long and doesn't converge well.

Answer:

Since the dataset contains only 500 images, training the entire ResNet50 is computationally expensive and can cause overfitting. So, freeze the pretrained layers and use ResNet50 as a feature extractor, training only the new classification head. If more accuracy is needed, then unfreeze only the last few layers and fine-tune them with a very small learning rate. Also using data augmentation and early stopping improves generalization.

Freeze the pretrained layers :

```
# Freeze ResNet50
```

```
base_model.trainable = False
```

Scenario 3: Transfer Learning Works on Training Data but Fails on New Environment

Context: You used a pretrained YOLOv8 model trained on outdoor images to detect tigers. But when used in night-mode CCTV footage, detection fails.

Problem: Domain shift between training data (bright daylight) and real-world application (night).

Answer:

1. Add night-time images to the training dataset Instead of training only on day time images
2. Simulating night-like conditions during training using augmentation
3. Check image quality in CCTV

Scenario 4: Object Detection Model Shows Bounding Boxes but Misses Object

Context: You trained a YOLOv5 model using transfer learning on a small vehicle dataset. Boxes are drawn, but not aligned with objects.

Problem: Bounding box coordinates are incorrect due to **wrong label format or image resizing** mismatch.

Answer:

1. verify the annotation format, YOLO expects class ID followed by normalized x-center, y-center, width and height values.
2. check whether the image resizing or annotation conversion has been performed consistently.
3. visualize the ground-truth bounding boxes on several training images before retraining. If the ground-truth boxes are already incorrect, should fix the annotations rather than changing the model.

Scenario 5: Model Gives High mAP on Training But Low Real-Time Accuracy

Context: You used pretrained YOLOv8 for multi-class object detection (helmet, vest, mask). mAP on test set is 90%, but real-time inference on webcam is inaccurate.

Problem: Difference in resolution, camera angle, or image quality between webcam and training data. Poor generalization due to **dataset bias**.

Answer:

- 1.This high mAP indicates the domain shift problem.
- 2.The training and test datasets may not represent the real webcam environment.
- 3.Should collect and annotate representative webcam images, including different resolutions, camera angles, lighting conditions, distances.
- 4.Should use suitable data augmentation and fine-tune the model on the new data.
- 5.Also evaluate the model on a separate validation set that closely represents the actual deployment environment.

Scenario 6: VGG16 Model Very Slow During Training

Project: Fashion product classification using **VGG16**

Problem:Model takes too long to train even with a small dataset and batch size.

Answer:

- 1.VGG16 is computationally expensive because it has a large number of parameters and many convolutional layers.
- 2.For a small fashion-product dataset, should use VGG16 with `include_top=False`, freeze the pretrained convolutional base `base_model.trainable = False`, and add a lightweight classification head.
- 3.Should initially train only the new layers.
- 4.If additional accuracy is required, can unfreeze the last few convolutional layers and fine-tune them using a very small learning rate.

Scenario 7: InceptionV3 Fails to Train Due to Input Size Mismatch

Project: Dog breed classification using **InceptionV3**

Problem:Runtime error due to input size mismatch.

Answer:

Should make the image preprocessing and model input dimensions consistent. For example, if I use the standard InceptionV3 configuration, should resize images to 299×299 RGB and set `input_shape=(299,299,3)`. Also verify the batch shape before training.